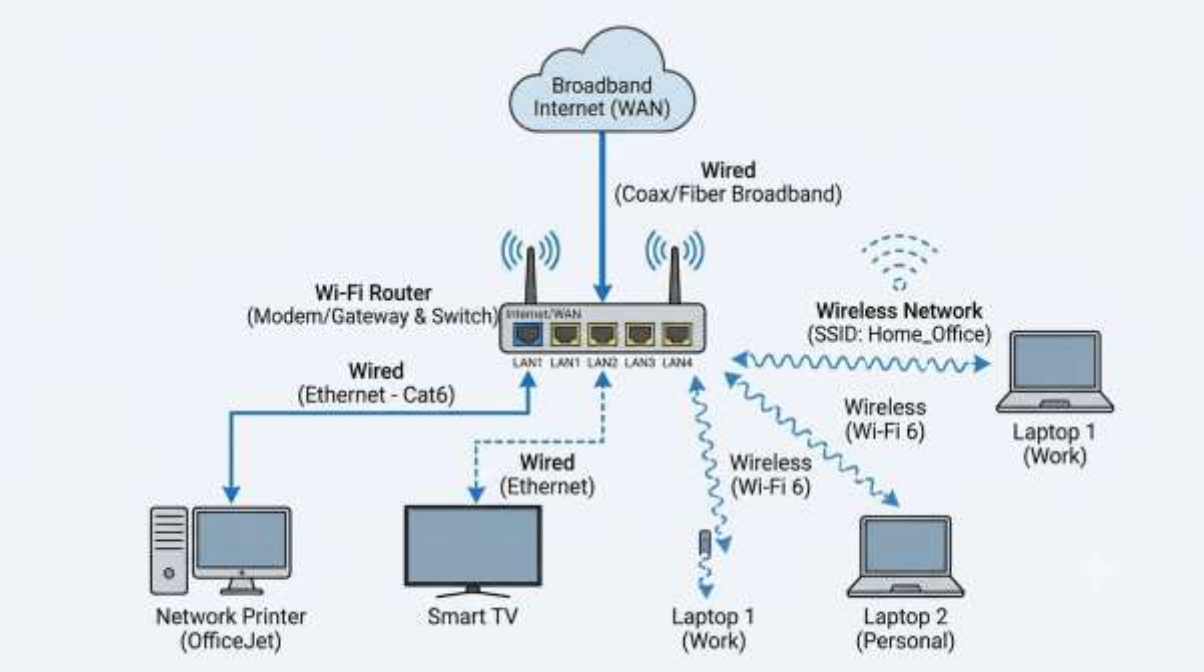
1.

Draw a network diagram for a small office/home office (SOHO) setup that includes: one broadband internet connection, one Wi-Fi router, two laptops, one smart TV, and one printer. Clearly label each device and connection type (wired or wireless).

ANS:-



2.

List all the hardware components (with brand/model suggestions) you would need to physically build the SOHO network diagrammed above, including cables and any switches or extenders if needed. Hint: Check current listings on Flipkart or Amazon for typical device models used in Indian homes/offices.

ANS:-

To build a reliable Small Office/Home Office (SOHO) network, the following hardware components with standard brand/model suggestions are recommended based on current availability:

- Wi-Fi Router: TP-Link Archer C6 (Dual-Band Gigabit Router) or D-Link R15 Wi-Fi 6 Router. Acts as the core gateway providing NAT, routing, and wireless coverage.
- Ethernet Switch: TP-Link LS1005 (5-Port) or D-Link DGS-1008A (8-Port Gigabit Switch). Used to expand wired LAN ports for stationary devices.
- Network Printer: HP LaserJet M126nw or Brother DCP-L2520D (Network/Wireless enabled printer).
- Smart TV / Display: Any standard Smart TV (e.g., Samsung 43-inch Crystal 4K or Xiaomi Smart TV) supporting Wi-Fi and Ethernet.
- Cabling & Accessories: Cat 6 UTP Ethernet Patch Cables (pre-terminated with RJ45 connectors) for wired runs between devices, router, and switch.

3.

Write step-by-step instructions to configure the Wi-Fi router for your SOHO network, including setting up a unique SSID, enabling WPA2 security, and changing the default admin password.

ANS:-

Step-by-Step Wi-Fi Router Configuration Guide

1.Connect to the Router and Access Admin Portal:Prerequisite.

Connect your laptop to the router via an Ethernet cable or the default Wi-Fi network printed on the router label. Open a web browser, enter the default gateway IP address (typically 192.168.0.1 or 192.168.1.1), and log in using the factory credentials.

Verification: The router's web management dashboard loads successfully in your browser.

2.Change Default Administrator Password:Security Best Practice.

Navigate to System Tools > Administration or Management settings. Enter a strong new administrator password to prevent unauthorized access to router settings.

Verification: You are prompted to re-authenticate with your new password when session times out.

3.Configure Unique SSID:Wireless Settings.

Go to the Wireless / Wi-Fi settings menu. Enter a unique network name (SSID)—for example, SOHO_Office_Net—for both the 2.4 GHz and 5 GHz bands.

Verification: The new SSID name appears in your computer's available Wi-Fi networks list.

4.Enable WPA2/WPA3 Security:Encryption.

Set the security mode to WPA2-PSK or WPA2/WPA3-Mixed. Enter a robust, secure Wi-Fi passphrase.

Verification: Connecting devices successfully authenticate and join the network using the new passphrase.

5.Save and Reboot:Finalization.

Click Save or Apply and reboot the router if prompted to apply all changes.

Verification: The router restarts, broadcasting the new SSID with active security encryption.

4.

Configure IP addresses for all devices in your SOHO network: assign static IPs to the printer and smart TV, and use DHCP for the laptops. Show a table listing each device, its MAC address (make one up), and the assigned IP address.

ANS:-

Static IP addresses are assigned to fixed infrastructure devices (Printer and Smart TV), while client devices (Laptops) obtain dynamic IP addresses via DHCP.

Device Name	MAC Address (Simulated)	Assignment Type	Assigned IP Address	Subnet Mask	Gateway
Office Printer	AA:BB:CC:11:22:33	Static	192.168.1.50	255.255.255.0	192.168.1.1
Smart TV	DD:EE:FF:44:55:66	Static	192.168.1.60	255.255.255.0	192.168.1.1
Laptop 1	11:22:33:AA:BB:CC	DHCP	192.168.1.10	255.255.255.0	192.168.1.1
Laptop 2	44:55:66:DD:EE:FF	DHCP	192.168.1.11	255.255.255.0	192.168.1.1
Laptop 3	77:88:99:00:11:22	DHCP	192.168.1.12	255.255.255.0	192.168.1.1

5.

Explain how you would troubleshoot if one of the laptops cannot connect to the internet in your SOHO network. List at least 3 possible causes and what steps you would take to diagnose and fix each issue.

ANS:-

If a laptop on your SOHO network cannot connect to the internet, follow these prioritized diagnostic steps:

Issue 1: IP Addressing / DHCP Failure (APIPA Address 169.x.x.x)

- Symptom: The laptop shows a self-assigned IP starting with 169.254.x.x and cannot reach local devices or the internet.
- Diagnostic Steps: Open Command Prompt and run ipconfig /all to check the assigned IPv4 address.
- Fix: Release and renew the lease by running ipconfig /release followed by ipconfig /renew. If it fails, check that the DHCP server service is active on the router and restart the network adapter.
- *Verification:* Running ipconfig shows a valid local IP address in the 192.168.1.x subnet.

Issue 2: Wi-Fi Authentication or Signal Association Failure

- Symptom: The laptop is connected to Wi-Fi with a "No Internet" warning yellow exclamation mark or fails to authenticate.
- Diagnostic Steps: Check the Wi-Fi icon status and verify if other devices on the same Wi-Fi network have internet access. Re-enter the Wi-Fi security key if authentication failed.
- Fix: Forget the network in Wi-Fi settings (Settings > Network & Internet > Wi-Fi > Manage Known Networks > Forget), then reconnect by entering the correct WPA2 password. Restart the laptop's Wi-Fi adapter.
- *Verification:* The Wi-Fi status shows "Connected, secured" and web pages load normally.

Issue 3: DNS Resolution Failure

- Symptom: You can ping external IP addresses (e.g., ping 8.8.8.8) successfully, but browsing websites by name (e.g., google.com) fails.
- Diagnostic Steps: Run nslookup google.com in Command Prompt to check if domain names are being resolved.
- Fix: Change the DNS server settings on the laptop network adapter to public DNS servers like Google (8.8.8.8) or Cloudflare (1.1.1.1). Clear the local DNS cache by running ipconfig /flushdns.
- *Verification:* nslookup returns a valid IP address for external domain names, and web browsers open sites instantly.